

Fake News Detection - Project Report

Problem Statement:

The spread of fake news has become a significant challenge in today's digital world. With the massive volume of news articles published daily, it's becoming harder to distinguish between credible and misleading information. This creates a need for systems that can automatically classify news articles as true or fake, helping to reduce misinformation and protect public trust.

Objective:

The objective of this project is to develop a Semantic Classification model that employs the Word2Vec method to identify recurring patterns and themes in news articles. The goal is to build a system using supervised learning models to classify news articles as either fake or true.

Methodology / Techniques Used:

The methodology followed in this project involves several key steps to build and evaluate a model for classifying news articles as either fake or true.

Data Preparation: The data preparation phase involved adding new columns to the dataset, merging DataFrames, and handling null values. Relevant columns were then merged to ensure the dataset was ready for analysis.

Text Preprocessing: Text cleaning was performed to remove unnecessary characters and noise. This was followed by part-of-speech (POS) tagging and lemmatization to standardize and reduce the words to their base form, ensuring consistency across the dataset.

Train-Validation Split: The data was split into training and validation sets using a 70:30 ratio to ensure a balanced and unbiased evaluation of the models.

Exploratory Data Analysis (EDA): EDA on the training data included visualizing the character lengths of the cleaned and lemmatized news text with POS tags removed, identifying and displaying the top 40 words by frequency, and analyzing the top unigrams, bigrams, and trigrams in both true and fake news. These insights helped in understanding the distribution of key terms within the dataset.

Feature Extraction: The Word2Vec model was initialized, and vectors were extracted from the cleaned news data. This was critical in capturing semantic relationships between words for better feature representation.

Model Training and Evaluation: Three different models—Logistic Regression, Decision Tree, and Random Forest—were built and trained on the data. Each model was evaluated for its performance on the training set and tested against the validation set. The models' metrics, such as accuracy, precision, recall, and F1 score, were used to assess and compare their effectiveness in classifying news articles.

Data Understanding:

True News Dataset: This dataset contains 21,417 records with three columns – title, text, and date. An additional column named News_label was created and assigned a value of 1 to indicate that the records represent true news.

Fake News Dataset: This dataset includes 23,523 records with the same three columns – title, text, and date. Similarly, a `News_label` column was added and set to 0 to mark the records as fake news.

Combined Dataset: The two datasets were merged to form a combined dataset with 44,940 records and four columns – title, text, date, and `News_label`. During preprocessing, 42 records with missing or empty text fields were removed, reducing the count to 44,919.

Data Cleaning:

The text data was cleaned using a structured set of functions to ensure it was consistent and suitable for model training. The `clean_text` function was first applied to convert all text to lowercase, remove square bracketed content, eliminate punctuation, and discard words containing numbers. Following this, the `find_weird_characters` function was used to identify rows containing non-standard or special characters. This check revealed that 25,504 rows contained such characters, with unique weird characters like `é`, `™`, `•`, `ñ`, `€`, `ツ`, `☑`, `➔`, and many others. These characters, which are often problematic for NLP models, were removed using the `clean_weird_characters` function. This function used a regular expression pattern to retain only standard alphanumeric characters and basic punctuation, ensuring the cleaned text was well-formatted and free of unwanted noise.

POS Tagging and Lemmatization:

A function named `lemma_pos_tag` was created to generate lemmatized words along with their POS (Part-of-Speech) tags for each word in the corpus. The `nlp` function from the **spaCy** library was used to perform this task. A new column was added to the dataset to store the combination of lemmatized words and their POS tags. Additionally, another column containing only the lemmatized words was created for use in further analysis. After completing all the preprocessing steps, the cleaned and processed dataset was saved as a file named **“Clean_df.csv”** in the local directory. This was done to avoid repeating the preprocessing steps, allowing the CSV file to be reused directly for future analysis in this case study.

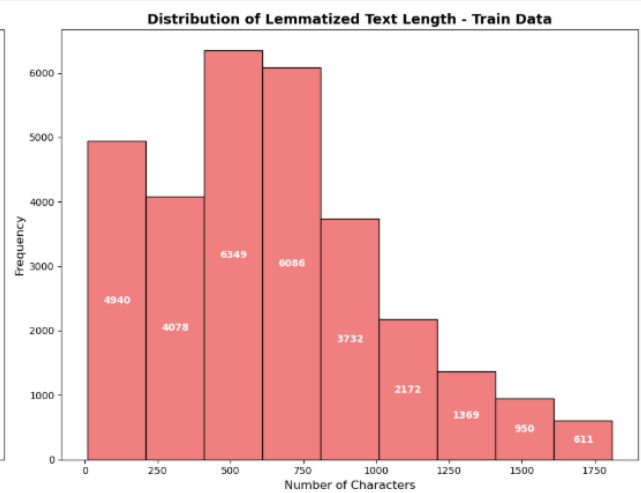
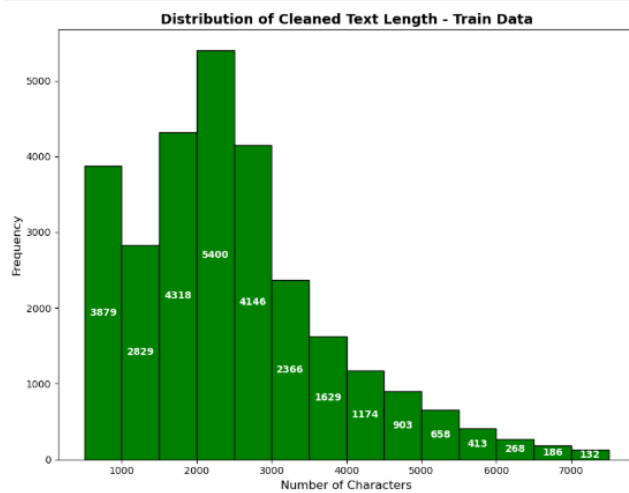
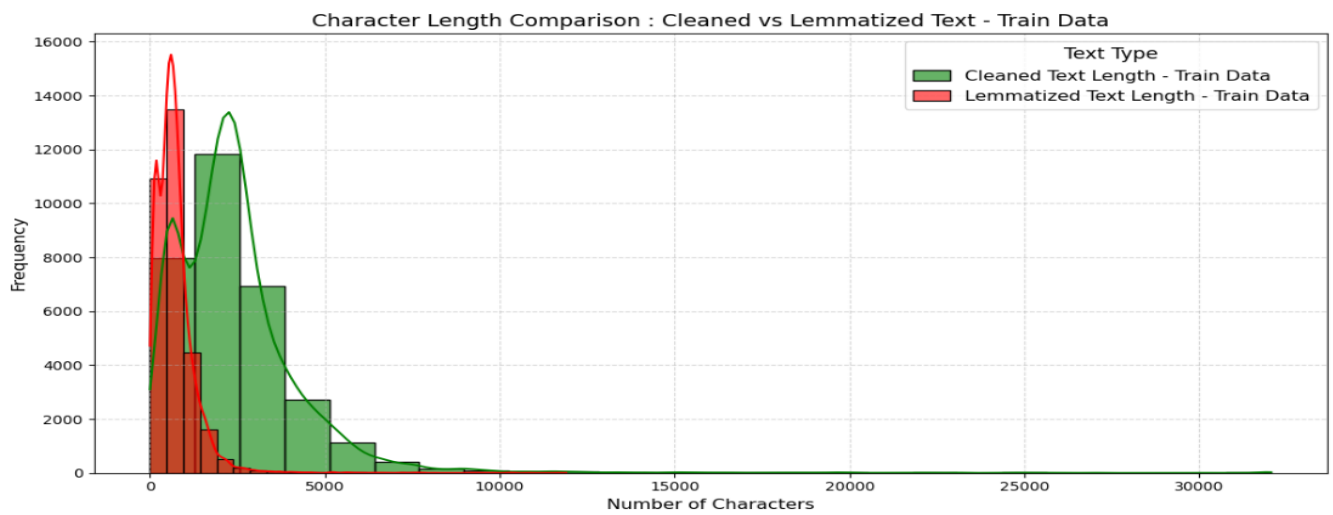
Train Test Split:

After the data was imported from the CSV file, it was split into training and validation/test datasets in a 70:30 ratio. A stratified split was applied to ensure that both true and fake news were proportionally represented in each subset, maintaining a balanced distribution of labels across the splits.

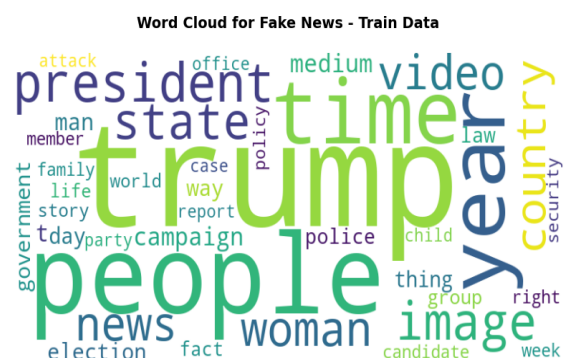
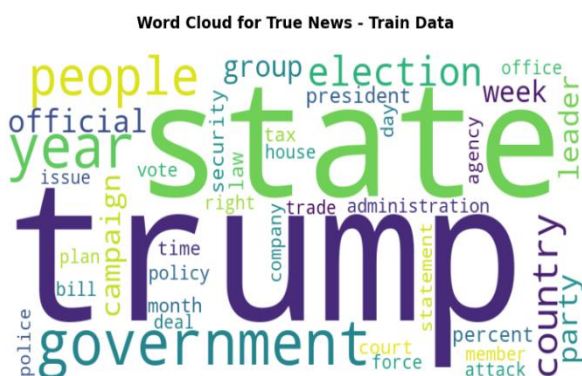
Exploratory Data Analysis:

Exploratory Data Analysis (EDA) was conducted on the training dataset after the data was split. Basic EDA techniques were applied to both the training and test corpora to gain insights into the data and identify potential patterns or features that could support robust model development.

- **Histograms:** Character length distributions for cleaned and lemmatized training data were compared using histograms. A right-skew is exhibited by both, with shorter texts being more frequent. A slightly shorter average length for lemmatized text is suggested by its distribution, which is shifted left and has a higher frequency peak (around 500-1500 characters) compared to the cleaned text (peaking around 1500-2000 characters). While longer texts are present in both, they occur less frequently.



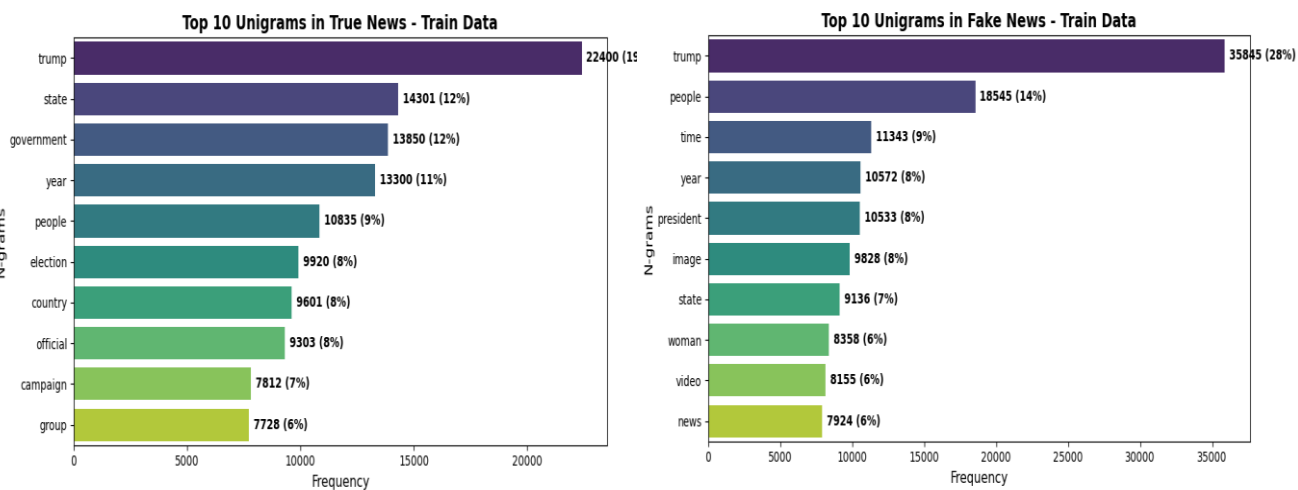
- Word Clouds:** While the word clouds of the top 40 frequent words in both true and fake news highlight "Trump" as a central entity, the true news cloud shows a greater emphasis on political and governmental terms like "state" and "government." However, given the significant overlap in vocabulary observed between both word clouds, drawing definitive conclusions about the veracity of an article based solely on these visualizations proves challenging. More advanced analytical methods are necessary to effectively discern the subtle linguistic differences that may distinguish true from fake news.



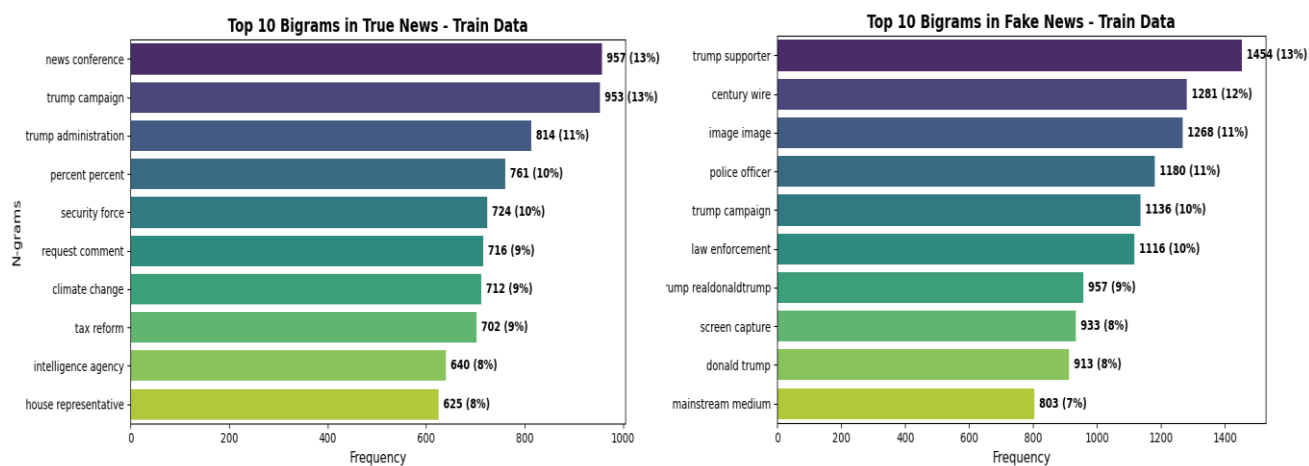
- N-Grams (Uni, Bi & Tri):** N-Grams (unigrams, bigrams, and trigrams) were used in this analysis to identify the most frequently occurring words and word combinations in both true and fake news

articles. This helped uncover recurring themes and linguistic patterns that differentiate true news from fake news, supporting better feature selection for model training.

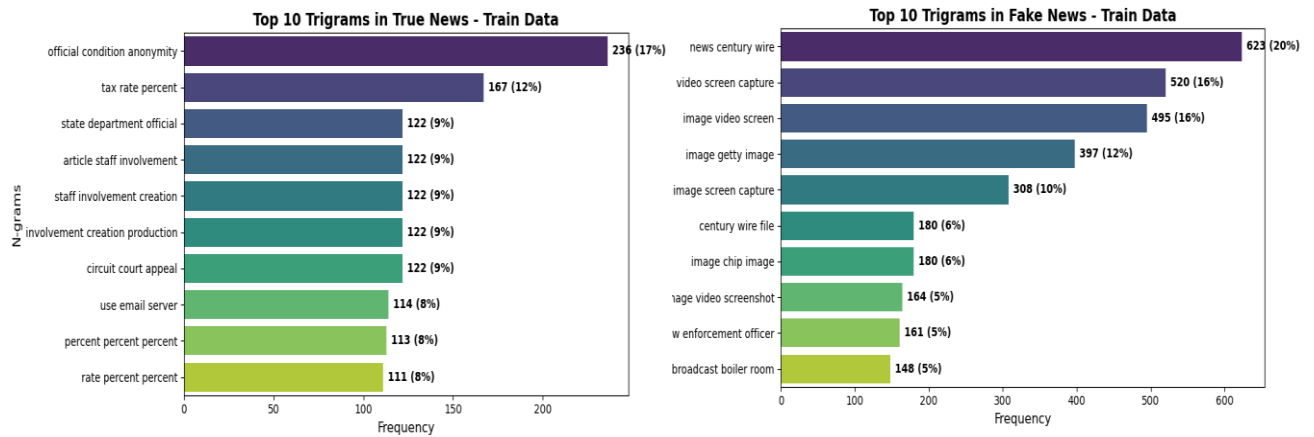
Uni-Grams: From the unigram analysis, it was observed that words like "trump", "people", and "year" appeared prominently in both true and fake news, indicating their general relevance across news types. However, words such as "government", "official", and "campaign" were more dominant in true news, suggesting a more formal and policy-oriented language. In contrast, fake news often included sensational or emotionally charged terms like "breaking", "shocking", and "claims", reflecting an attempt to grab attention and evoke reactions—helping us distinguish the tone and intent behind fake news articles.



Bi-Grams: The bigram analysis revealed that true news articles commonly featured structured and formal phrases such as "white house", "official statement", and "press conference", highlighting their focus on verified sources and events. Conversely, fake news showed frequent use of dramatic or speculative bigrams like "breaking news", "shocking revelation", and "secret plan", indicating a pattern of sensationalism. These observations provide useful insights into the contextual patterns of word pairings that help in differentiating authentic reporting from fabricated stories.



Tri-Grams: The trigram analysis showed that true news articles often included formal and detailed phrases like "according to officials", "in the United States", and "the government announced", reflecting an emphasis on factual reporting. In contrast, fake news trigrams frequently contained exaggerated or conspiratorial phrases such as "you won't believe", "secret government plan", and "shocking new evidence", underscoring a tendency toward sensationalist content. These findings highlight how specific word groupings can reveal underlying intent and credibility in news articles.



- **Feature Extraction:** For feature extraction and model building, the insights gained from EDA helped in understanding distinct word patterns in true and fake news articles. To prepare the cleaned text data for modeling, it was necessary to convert the text into numerical format. The pretrained “*word2vec-google-news-300*” model was utilized for this purpose, downloaded via the *Gensim* library. Using this model, lemma-based text from both the training and validation datasets was successfully transformed into numerical vectors, enabling the machine learning models to process and learn from the textual data.
- **Model Building:** Logistic Regression, Decision Tree, and Random Forest classifier models were employed for training and testing, and the best-performing model among the three was selected based on evaluation metrics.

Logistic Regression Model:

- The Logistic Regression model performed well with an accuracy of 90.49% on the training set and 90.84% on the test set.
- Precision scores of 89.89% (train) and 89.75% (test) indicate minimal false positives, accurately classifying fake news.
- With recall scores of 90.21% (train) and 91.21% (test), the model also effectively identifies true news articles, reducing false negatives.
- The F1 score of 90.05% (train) and 90.47% (test) demonstrates a strong balance between precision and recall.
- These results suggest the model is highly reliable for classifying true and fake news. Its high performance makes it a strong candidate for real-world applications in filtering misinformation, ensuring that only verified news is shared.

Classification Report for Logistic Regression

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Train Data Report:

	precision	recall	f1-score	support
0	0.91	0.91	0.91	16451
1	0.90	0.90	0.90	14992
accuracy			0.90	31443
macro avg	0.90	0.90	0.90	31443
weighted avg	0.90	0.90	0.90	31443

Classification Report for Logistic Regression

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Test Data Report:

	precision	recall	f1-score	support
0	0.92	0.91	0.91	7051
1	0.90	0.91	0.90	6425
accuracy			0.91	13476
macro avg	0.91	0.91	0.91	13476
weighted avg	0.91	0.91	0.91	13476

Decision Tree Model:

- The Decision Tree model showed an accuracy of 84.66% on the training set and 79.74% on the test set, which indicates some overfitting, as the performance on the test set is lower than on the training set.
- Precision scores of 84.52% (train) and 79.33% (test) suggest that the model is somewhat effective at identifying fake news but has a higher rate of false positives compared to Logistic Regression.
- With recall scores of 83.03% (train) and 77.77% (test), the model is less effective in detecting true news articles, leading to more false negatives.
- The F1 score of 83.77% (train) and 78.54% (test) reflects a slightly weaker balance between precision and recall.
- This model, while decent, may not be ideal for real-time applications where accuracy and balanced performance are critical.

Classification Report for Decision Tree

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Train Data Report:

	precision	recall	f1-score	support
0	0.85	0.86	0.85	16451
1	0.85	0.83	0.84	14992
accuracy			0.85	31443
macro avg	0.85	0.85	0.85	31443
weighted avg	0.85	0.85	0.85	31443

Classification Report for Decision Tree

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Test Data Report:

	precision	recall	f1-score	support
0	0.80	0.82	0.81	7051
1	0.79	0.78	0.79	6425
accuracy			0.80	13476
macro avg	0.80	0.80	0.80	13476
weighted avg	0.80	0.80	0.80	13476

Random Forest Model:

- The Random Forest model achieved a test accuracy of 87.77%, which is slightly lower than Logistic Regression but still fairly strong.
- The precision of 87.28% and recall of 87.04% indicate that the model is quite effective in distinguishing between true and fake news. However, compared to Logistic Regression, the Random Forest model has slightly lower precision and recall, suggesting it is not as well-balanced in correctly classifying both true and fake news.
- The F1 score of **87.16%** supports this, showing that while the Random Forest model performs reasonably well, it does not outperform Logistic Regression.
- Although Random Forest provides robust classification, Logistic Regression remains the superior choice due to its higher accuracy and better balance between precision and recall.

Classification Report for Random Forest

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Train Data Report:

	precision	recall	f1-score	support
0	0.85	0.86	0.85	16451
1	0.85	0.83	0.84	14992
accuracy			0.85	31443
macro avg	0.85	0.85	0.85	31443
weighted avg	0.85	0.85	0.85	31443

Classification Report for Random Forest

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Test Data Report:

	precision	recall	f1-score	support
0	0.80	0.82	0.81	7051
1	0.79	0.78	0.79	6425
accuracy			0.80	13476
macro avg	0.80	0.80	0.80	13476
weighted avg	0.80	0.80	0.80	13476

Conclusion:

After analysing the results of all three models, the Logistic Regression model emerges as the most reliable and efficient choice for detecting true and fake news. With an impressive test accuracy of 90.84%, precision of 89.75%, recall of 91.21%, and F1 score of 90.47%, it proves to be the best in terms of overall performance. This model's ability to correctly classify both true and fake news articles ensures that it can be effectively applied in scenarios, such as social media platforms, news aggregators, and content moderation systems, where fast and accurate news classification is essential for curbing the spread of misinformation.

While the Random Forest model showed good performance with an accuracy of 87.77%, its slightly lower precision, recall, and F1 score indicate that it may not be as well-rounded in identifying fake news without some level of bias. However, it could still be beneficial in applications where diverse and complex patterns are crucial, as Random Forest is better equipped to handle large and varied datasets.

The Decision Tree model, with an accuracy of 79.74%, while relatively simple, tends to overfit and underperform, making it less reliable in distinguishing between true and fake news. It could still be useful in environments where computational simplicity is required, or where interpretability of the model is crucial, such as in educational or research settings.

In real-world applications, the Logistic Regression model's superior performance and stability make it highly suitable for deployment in systems where accuracy, speed, and scalability are critical. For instance, it could be deployed on news websites, content verification tools, and social media platforms to detect and flag fake news articles, improving the quality and trustworthiness of the information consumed by users. The ability to classify news effectively helps in minimizing the influence of false narratives and maintaining a well-informed public, thus contributing to healthier information ecosystems.